Defining Quality

- Definition of quality is dependent on the people defining it
- There is no single, universal definition of quality
- 5 common definitions include:

Defining Quality

1. Conformance to specifications
   - Does product/service meet targets and tolerances defined by designers?
2. Fitness for use
   - Evaluates performance for intended use
3. Value for price paid
   - Evaluation of usefulness vs. price paid
4. Support services
   - Quality of support after sale
5. Psychological
   - Ambience, prestige, friendly staff

Eight dimensions of quality：David A. Garvin, Harvard

Dimension	Meaning / Definition	Example	Focus Area
1. Performance	Primary operating characteristics of a product—how well it does what it is supposed to do.	Speed of a laptop, fuel efficiency of a car.	Functionality and measurable output.
2. Features	Secondary characteristics that add to product appeal or convenience.	Touchscreen in a refrigerator, GPS in a car.	Added functionalities or extras.
3. Reliability	Probability of performing intended function without failure over a specified period.	LED bulb that works 10,000 hours without flicker.	Consistency and dependability.
4. Conformance	Degree to which product meets established standards or specifications.	Tablets meeting FDA manufacturing norms.	Adherence to design/production standards.

Dimension	Meaning / Definition	Example	Focus Area
5. Durability	The amount of use before product deteriorates or requires replacement.	Tyres lasting 60,000 km before wear-out.	Lifespan and wear resistance.
6. Serviceability	Ease, speed, and cost of repair or maintenance.	Car with quick, affordable servicing.	After-sales support and maintainability.
7. Aesthetics	How the product looks, feels, sounds, tastes, or smells.	Elegant smartphone design, luxury car interior.	Sensory appeal and design.
8. Perceived Quality	Customer's perception of overall quality based on brand reputation, advertising, or prior experience.	Apple's perceived premium quality.	Brand image and customer trust.

Manufacturing Quality vs. Service Quality

- Manufacturing quality focuses on tangible product features
  - Conformance, performance, reliability, features
- Service organizations produce intangible products that must be experienced
  - Quality often defined by perceptional factors like courtesy, friendliness, promptness, waiting time, consistency

Cost of Quality

- Quality affects all aspects of the organization
- Quality has dramatic cost implications:
  - Quality control costs (Cost of conformance)
    - Prevention costs: planning, documenting, process control, training, consulting, pilot runs
    - Appraisal costs: measuring, evaluating, auditing, tests and inspection
  - Quality failure costs (Cost of non-conformance)
    - Internal failure costs: scrap, rework, repair
    - External failure costs: complaints, service, returns, concessions

Statistical Process Control

- Collective set of tools & techniques used to develop a quality assurance system when business processes exhibit variations is known as Statistical Process Control (SPC)
- Key issues addressed in SPC based quality assurance system:
  - How does one ensure that the random events are indeed rare events?
  - How do we know whether the observed changes are due to random variations or assignable causes?

Common & Assignable Causes

- Two types of variations occur in business processes
  - Chance variations due to **common causes**
    - Common causes are those due to random events that cannot be controlled
  - Non-random variations due to **assignable causes**
    - When observed variations are not statistically found to be due to random events, it clearly points to the existence of assignable causes
      - Errors due to operator skill level differences
      - Changes in the operating condition of an equipment
      - Changes introduced in the standard operating procedures
- In the case of assignable causes, the system will drift from desired level of performance over time

Quality Assurance using SPC

Some terminologies

Designed Standard

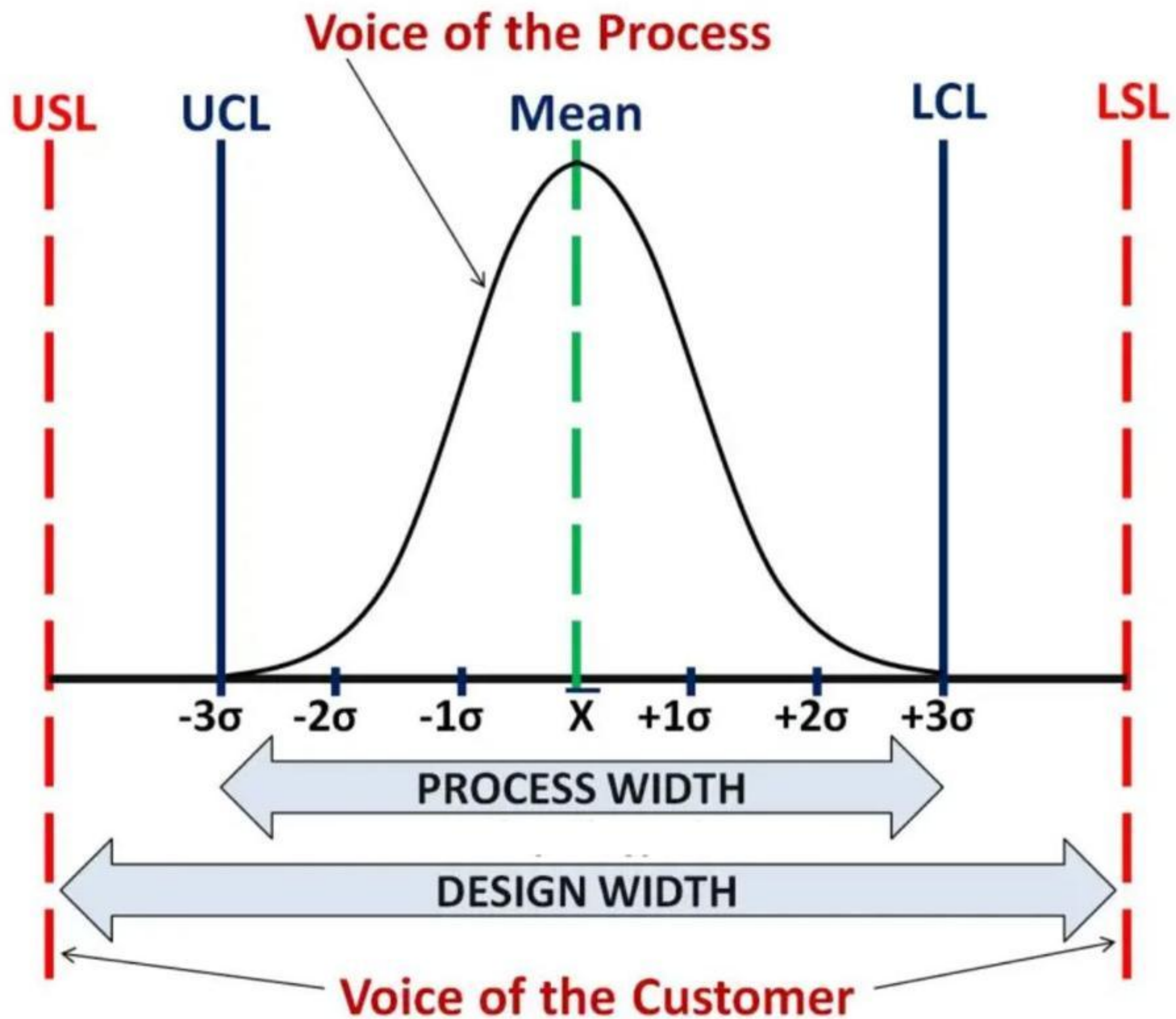
- Centre of specification limits (Target)
- Upper Specification Limit (USL)
- Lower Specification Limit (LSL)
- (USL – LSL): Desired Tolerance for the specs.

This represents the **voice of the customer**

Status of process

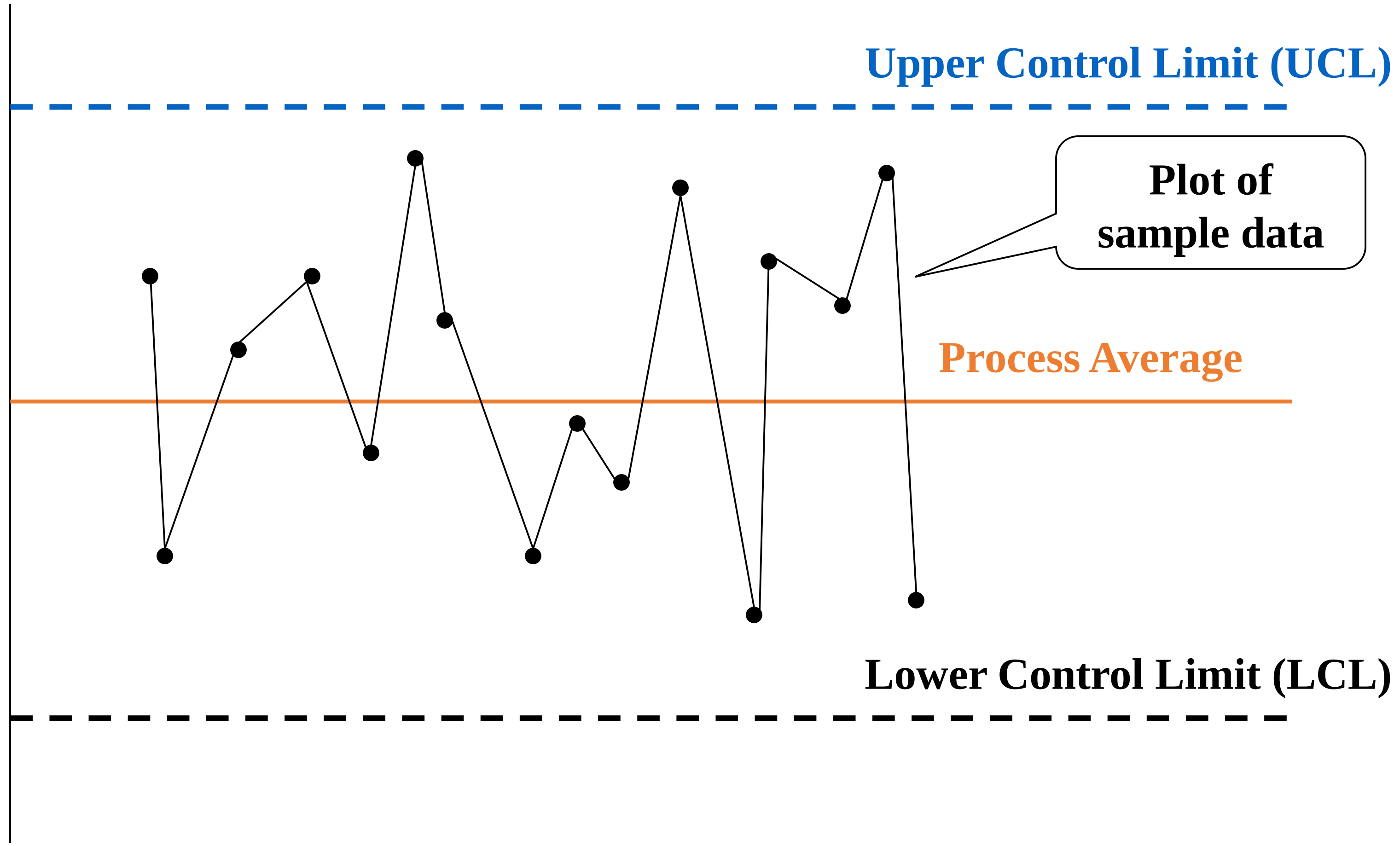
- Centre of the process (Process Average)
- Upper Control Limit (UCL)
- Lower Control Limit (LCL)
- (UCL – LCL): Spread of the process

This represents the **voice of the process**

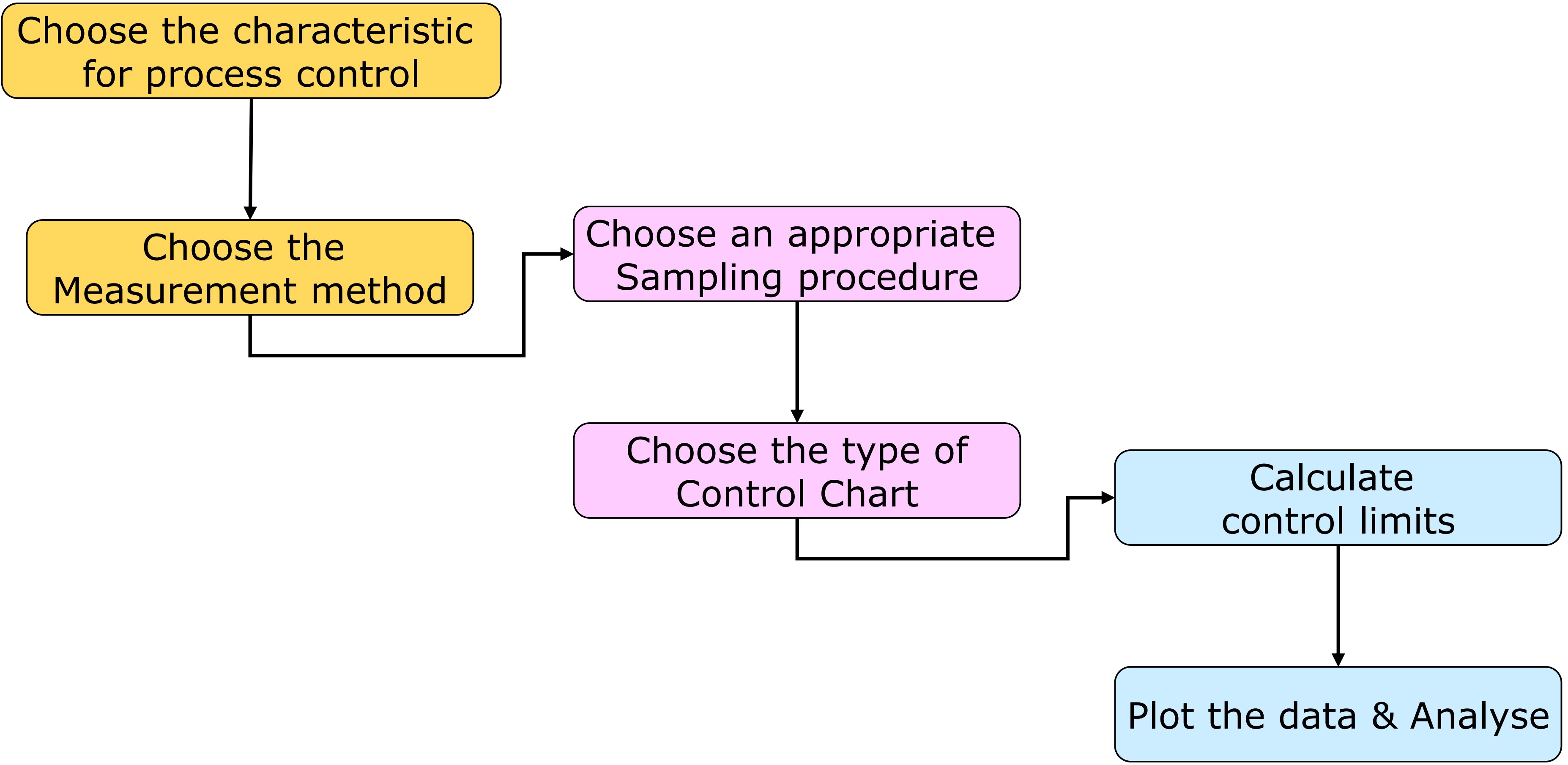


Voice of the Process
USL
UCL
Mean
LCL
LSL
-3σ
-2σ
-1σ
X̄
+1σ
+2σ
+3σ
PROCESS WIDTH
DESIGN WIDTH
Voice of the Customer

Control Chart
A generalized representation



Setting up a process control system



Characteristics for process control
Some examples

Sl. No.	Type of Applications	Characteristic for Measurement
1	Component Manufacturing	▪ Conformance of physical measurements of components and sub-assemblies to specifications ▪ Conformance to operating characteristics of machines and other resources involved in the process
2	Final Assembly	▪ Number of defects in the product ▪ Conformance to test specifications ▪ Number of missing elements
3	Process Industries	▪ Temperature, Pressure and Heat specifications ▪ Conformance to product specifications ▪ Conformance to equipment specifications ▪ Vibrations and other variations in equipments and sub-systems ▪ Conformance to specifications of the automation & control system
4	Service Systems	▪ Number of defects in various business processes ▪ Errors in processing documents ▪ Conformance to waiting time/lead time related specifications

Choosing a characteristic
Examples from service industry

- How long does it take to complete the patient admission process in a hospital?
- How dissatisfied were the customers when they stayed in a hotel for a holiday?
- Does our process of understanding customer requirements need improvement?
- How capable are my sub-processes in software development?

Measurement Methods

- **Attribute Based**
  - simple clustering of the characteristic into a few categories (such as good or bad)
  - Two frequently used attribute measures are:
    - Proportion of defects (denoted as p)
    - Number of defects (denoted as c)
  - measurements are easy to make, quick & less expensive, but reveal very little information about the process
- **Variable Based**
  - Detailed observation of the characteristic (such as length, diameter, weight)
  - measurement will be expensive and more time consuming but will provide a wealth of information about the process

Types of Control Charts

- Variable based control chart
  - X bar and R Charts
- Attribute based control chart
  - P chart
  - C chart

Central Limit Theorem

Regardless of the distribution of the population, the distribution of sample means drawn from the population will tend to follow a normal curve

1) The mean of the sampling distribution will be the same as the population mean μ

$$\bar{\bar{x}} = \mu$$

2) The standard deviation of the sampling distribution $(\sigma_{\bar{x}})$ will equal the population standard deviation (σ) divided by the square root of the sample size, n

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

X-bar chart : Setting Chart Limits

For $\bar{x}$-Charts when we know σ

$$\text{Upper control limit (UCL)} = \bar{\bar{x}} + z\sigma_{\bar{x}} = \bar{\bar{x}} + z\frac{\sigma}{\sqrt{n}}$$

$$\text{Lower control limit (LCL)} = \bar{\bar{x}} - z\sigma_{\bar{x}} = \bar{\bar{x}} - z\frac{\sigma}{\sqrt{n}}$$

Where

$\bar{\bar{x}} =$ mean of the sample means or a target value set for the process

$z =$ number of normal standard deviations

$\sigma_{x} =$ standard deviation of the sample means $= \frac{\sigma}{\sqrt{n}}$

$\sigma =$ population (process) standard deviation

$n =$ sample size

Setting Control Limits

- Randomly select and weigh nine $(n = 9)$ boxes each hour

Average weight in the first sample $= \dfrac{17+13+16+18+17+16+15+17+16}{9} = 16.1$ ounces

Weight of Sample		Weight of Sample		Weight of Sample	
Hour	**(Average of 9 Boxes)**	**Hour**	**(Average of 9 Boxes)**	**Hour**	**(Average of 9 Boxes)**
1	16.1	5	16.5	9	16.3
2	16.8	6	16.4	10	14.8
3	15.5	7	15.2	11	14.2
4	16.5	8	16.4	12	17.3

Setting Control Limits

Average mean of 12 samples $= \left[\bar{\bar{x}} = \dfrac{\sum_{i=1}^{12} (\text{Avg of 9 boxes})}{12} \right]$

$\bar{\bar{x}} = 16 \text{ ounces}$

$n = 9$

$z = 3$

$\sigma = 1 \text{ ounce}$

Number of samples $= 12$

Setting Control Limits

Average mean of 12 samples

$$= \left[\overline{\overline{x}} = \frac{\sum_{i=1}^{12} (\text{Avg of 9 boxes})}{12} \right]$$

$\overline{\overline{x}} = 16$ ounces

$n = 9$

$z = 3$

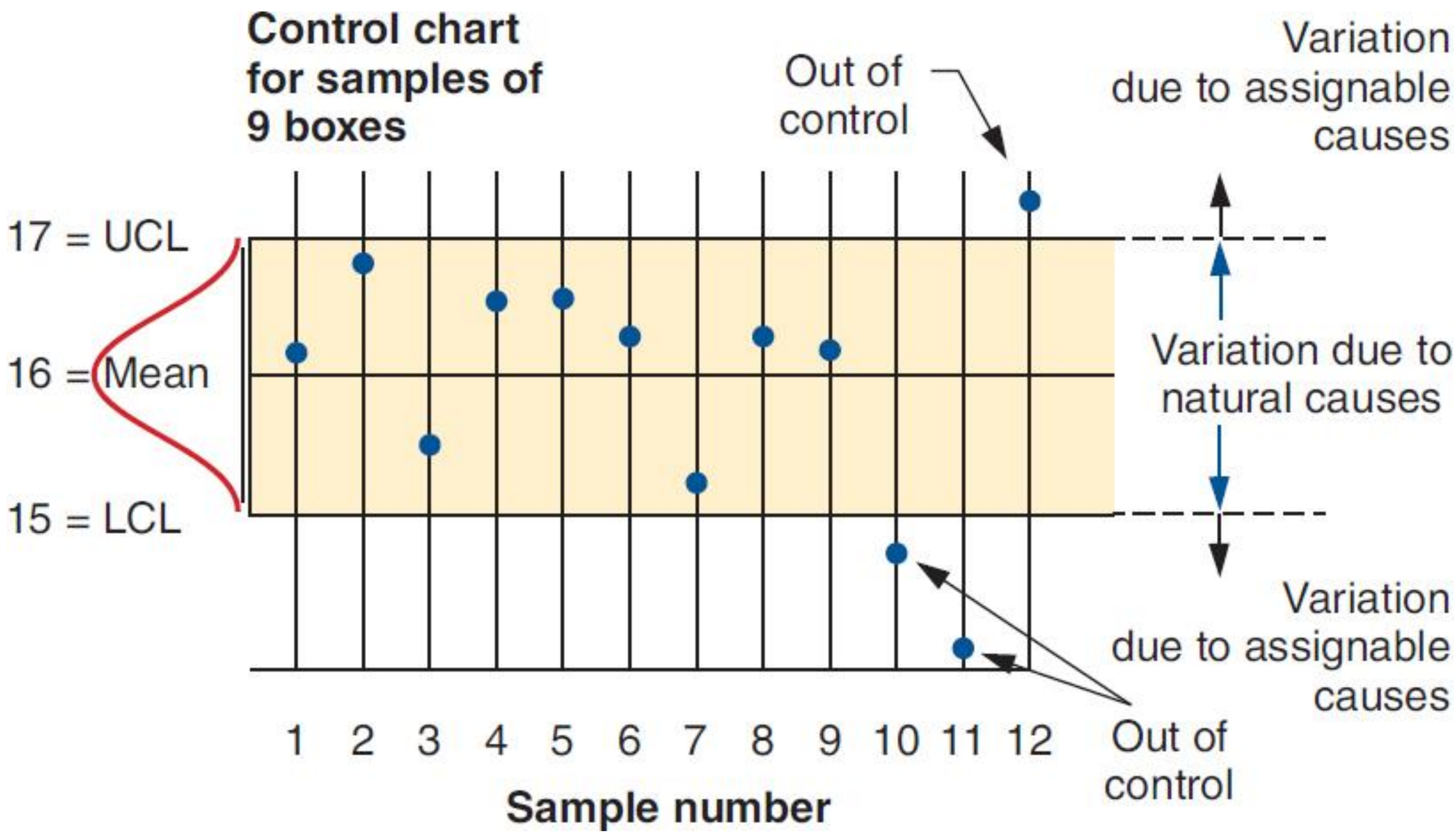
$\sigma = 1$ ounce

Number of samples $= 12$

$$\text{UCL}_{\overline{x}} = \overline{\overline{x}} + z\sigma_{\overline{x}} = 16 + 3\left(\frac{1}{\sqrt{9}}\right) = 16 + 3\left(\frac{1}{3}\right) = 17 \text{ ounces}$$

$$\text{LCL}_{\overline{x}} = \overline{\overline{x}} - z\sigma_{\overline{x}} = 16 - 3\left(\frac{1}{\sqrt{9}}\right) = 16 - 3\left(\frac{1}{3}\right) = 15 \text{ ounces}$$

Setting Control Limits



X-bar chart : Setting Chart Limits

For $\bar{X}$ - Charts when we don't know σ Here σ is estimated as, $\sigma = \frac{\bar{R}}{d_2}$

$$\text{Upper control limit (UCL)} = \bar{\bar{x}} + z\sigma_{\bar{x}} = \bar{\bar{x}} + z\frac{\sigma}{\sqrt{n}} = \bar{\bar{x}} + z\frac{\bar{R}}{d_2\sqrt{n}}$$

$$\text{Lower control limit (LCL)} = \bar{\bar{x}} - z\sigma_{\bar{x}} = \bar{\bar{x}} - z\frac{\sigma}{\sqrt{n}} = \bar{\bar{x}} - z\frac{\bar{R}}{d_2\sqrt{n}}$$

if z is taken as 3, $UCL_{\bar{x}} = \bar{x} + A_2\bar{R}$ $\quad A_2 = \frac{3}{d_2\sqrt{n}}$

$LCL_{\bar{x}} = \bar{x} - A_2\bar{R}$

where $\bar{R} = \frac{\sum_{i=1}^{k} R_i}{k} =$ average range of the samples

d_2 *and* A_2 = control chart factor found in table,

$\bar{\bar{x}}$ = mean of the sample means

R_i = range for sample *i*

k = total number of samples

Coefficients for computing LCL and UCL in X-bar and R charts*

Sample size (*n*)	A_2	D_3	D_4
2	1.880	0	3.268
3	1.023	0	2.574
4	0.729	0	2.282
5	0.577	0	2.114
6	0.483	0	2.004
7	0.419	0.076	1.924
8	0.373	0.136	1.864
9	0.337	0.184	1.816
10	0.308	0.223	1.777

Setting Control Limits

Soft drink bottles labeled as “net weight 12 ounces”

Process average = 12 ounces

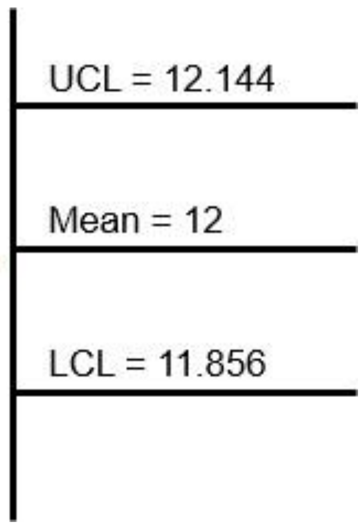
Average range = .25 ounces

Sample size = 5

$$
\begin{aligned}
UCL_{\bar{x}} &= \bar{\bar{x}} + A_2\bar{R} \\
&= 12 + (.577)(.25) \\
&= 12 + .144 \\
&= 12.144 \text{ ounces}
\end{aligned}
$$

From Table

$$
\begin{aligned}
LCL_{\bar{x}} &= \bar{\bar{x}} - A_2\bar{R} \\
&= 12 - .144 \\
&= 11.856 \text{ ounces}
\end{aligned}
$$



R – Chart

- Type of variables control chart
- Shows sample ranges over time
  - Difference between smallest and largest values in sample
- Monitors process variability
- Independent from process mean

Setting Chart Limits

For *R*-Charts

$$\text{Upper control limit } (\text{UCL}_R) = D_4\bar{R}$$

$$\text{Lower control limit } (\text{LCL}_R) = D_3\bar{R}$$

where

UCL_R = upper control limit for the range

LCL_R = lower control limit for the range

D_4 and D_3 = values from Table

Setting Control Limits

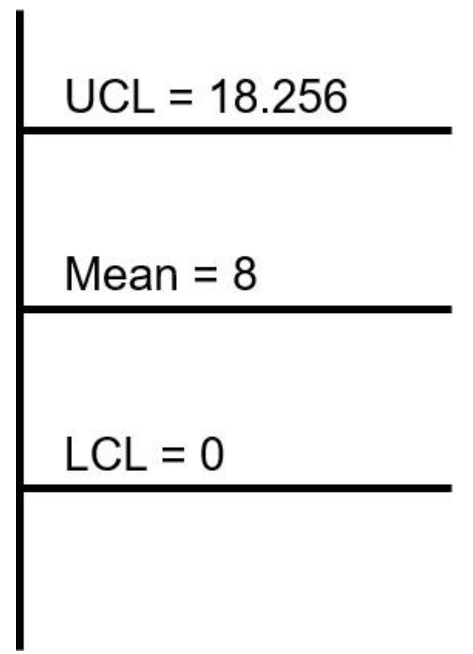
Average range = 8 minutes

Sample size = 4

From Table , $D_4 = 2.282, \quad D_3 = 0$

$$\begin{aligned} UCL_R &= D_4\bar{R} \\ &= (2.282)(8) \\ &= 18.256 \text{ minutes} \end{aligned}$$

$$\begin{aligned} LCL_R &= D_3\bar{R} \\ &= (0)(8) \\ &= 0 \text{ minutes} \end{aligned}$$



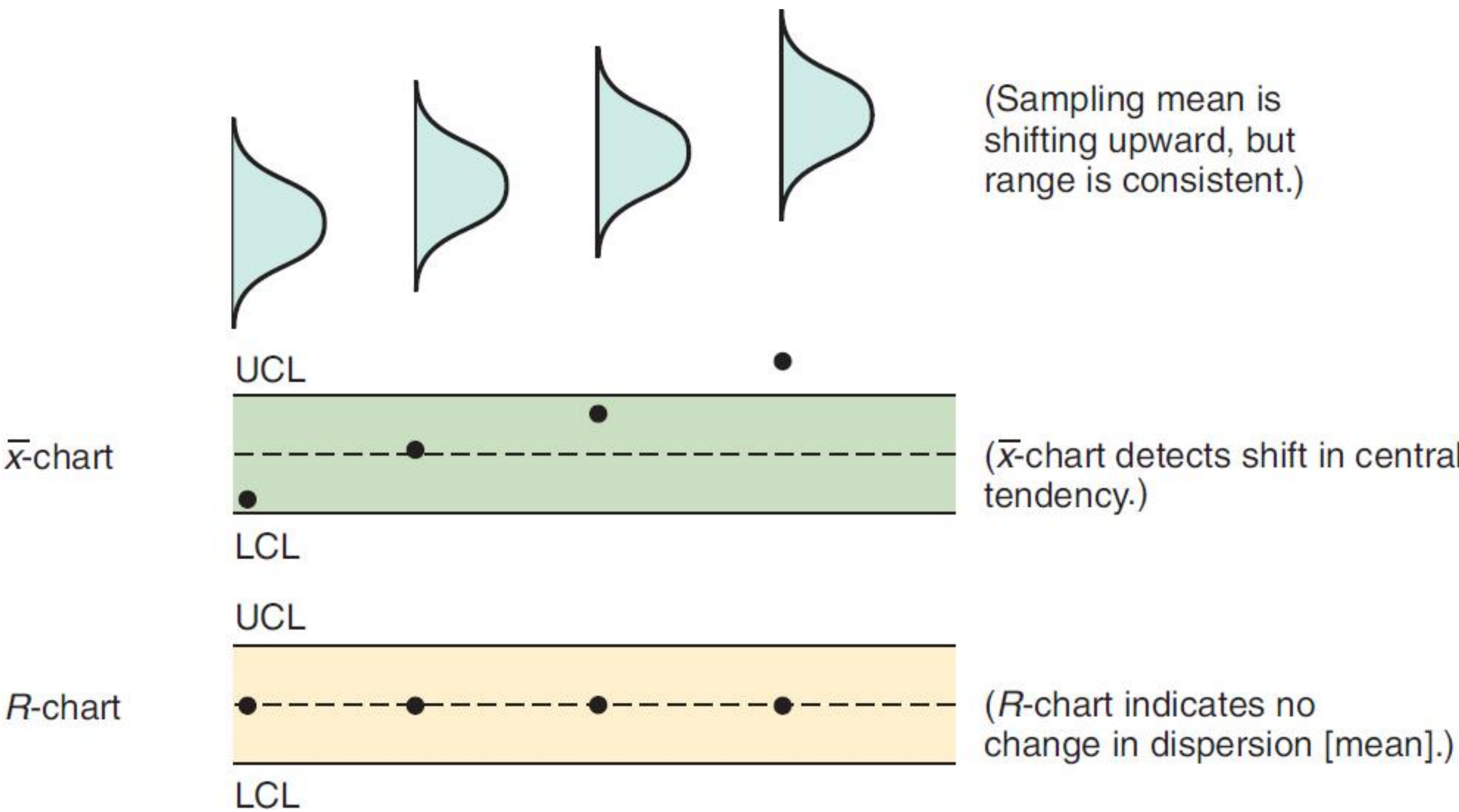
Example

Example 18.1. The following data were obtained over a five-day period to indicate $\bar{X}$ and R control chart for a quality characteristic of a certain manufacturing product that had required a substantial amount of rework. All the figures apply to the product made on a single machine by a single operator. The sample size was 5. Two samples were taken per day. Comment on the process using $\bar{X}$- and R-charts.

Sample Number	Observations					$\bar{X}$	R
	1	2	3	4	5		
1	10	12	13	8	9	10.4	5
2	7	10	8	11	9	9.0	4
3	11	12	9	12	10	10.8	3
4	10	9	8	13	11	10.2	5
5	8	11	11	7	7	8.8	4
6	11	8	8	11	10	9.6	3
7	10	12	13	13	9	11.4	4
8	10	12	12	10	12	11.2	2
9	12	13	11	12	10	11.6	3
10	10	13	7	9	12	10.2	6

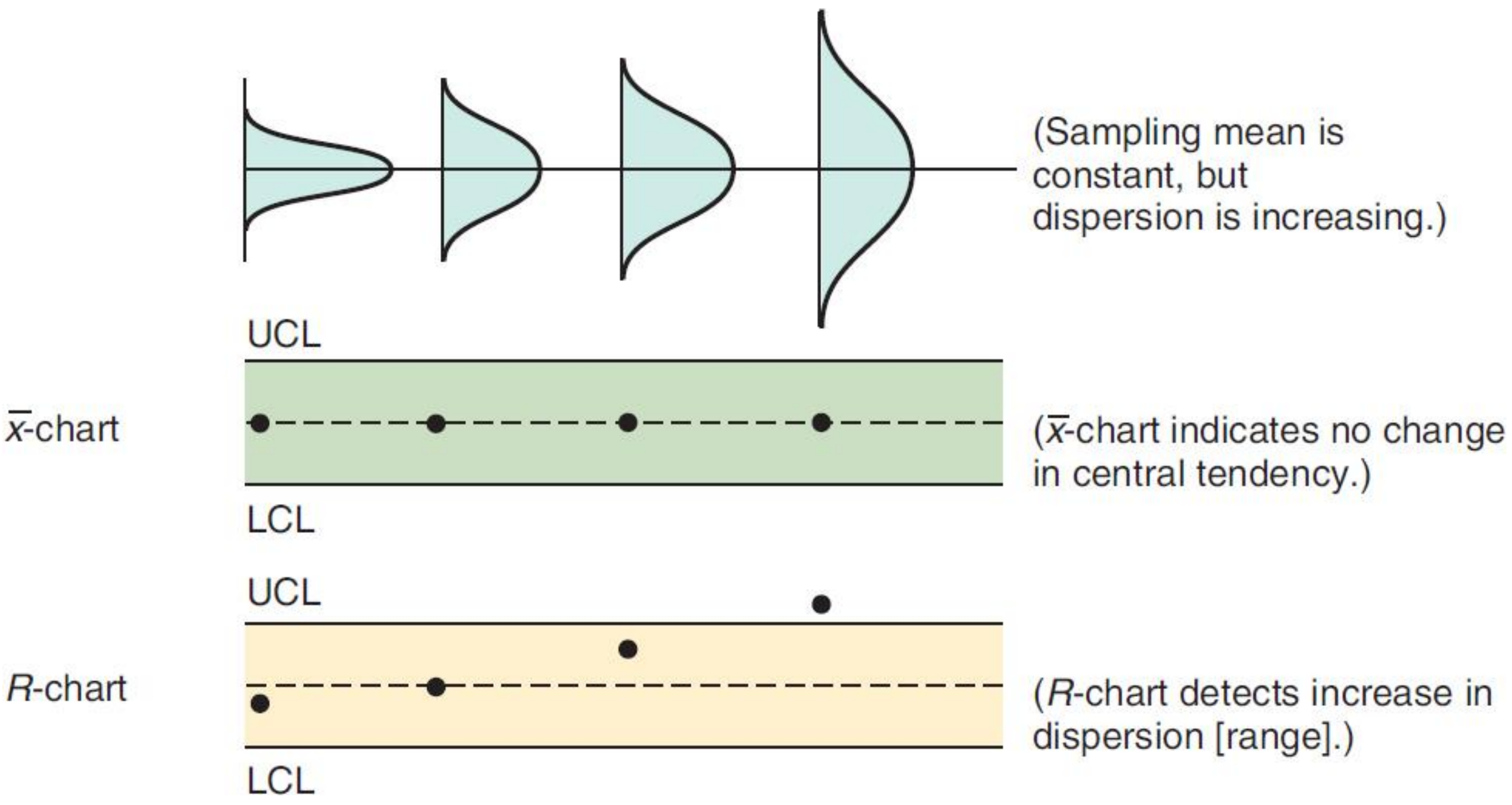
Mean and Range Charts

(a) These sampling distributions result in the charts below



Mean and Range Charts

(b) These sampling distributions result in the charts below



Steps in Building Control Charts

1. Collect 20 to 25 samples, often of n = 4 or n = 5 observations each, from a stable process, and compute the mean and range of each

2. Compute the overall means $(\bar{\bar{x}}$ and $\bar{R})$, set appropriate control limits, usually at the 99.73% level, and calculate the preliminary upper and lower control limits

- **If the process is not currently stable and in control**, use the desired mean, μ, instead of $\bar{\bar{X}}$ to calculate limits.

Steps in Creating Control Charts

3. Graph the sample means and ranges on their respective control charts and determine whether they fall outside the acceptable limits
4. Investigate points or patterns that indicate the process is out of control – try to assign causes for the variation, address the causes, and then resume the process
5. Collect additional samples and, if necessary, revalidate the control limits using the new data

Setting Other Control Limits

Common *z* Values

Desired Control Limit (%)	Z-Value (Standard Deviation Required for Desired Level of Confidence)
90.0	1.65
95.0	1.96
95.45	2.00
99.0	2.58
99.73	3.00

Control Charts for Attributes

- For variables that are categorical
  - **Defective/ nondefective**, good/ bad, yes/no, acceptable/unacceptable
- Measurement is typically counting defectives
- Charts may measure

1. **Percent** defective (*p*-chart)
2. **Number** of defects (*c*-chart)

Control Limits for *p*-Charts

Population will be a binomial distribution, but applying the central limit theorem allows us to assume a normal distribution for the sample statistics

$$\text{UCL}_p = \bar{p} + z\sigma_p$$

$$\text{LCL}_p = \bar{p} - z\sigma_p$$

σ_p is estimated by

$$\hat{\sigma}_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

where

$\bar{p} =$ mean fraction (percent) defective in the samples

$z =$ number of standard deviations

$\sigma_p =$ standard deviation of the sampling distribution

$n =$ number of observation in **each** sample

Example: p-Chart for Data Entry

100 records entered by each clerk are examined

Sample Number	Number Of Errors	Fraction Defective	Sample Number	Number Of Errors	Fraction Defective
1	6	.06	11	6	.06
2	5	.05	12	1	.01
3	0	.00	13	8	.08
4	1	.01	14	7	.07
5	4	.04	15	5	.05
6	2	.02	16	4	.04
7	5	.05	17	11	.11
8	3	.03	18	3	.03
9	3	.03	19	0	.00
10	2	.02	20	4	.04
				80	

p-Chart for Data Entry

$$\bar{p} = \frac{\text{Total number of errors}}{\text{Total number of records examined}} = \frac{80}{(100)(20)} = .04$$

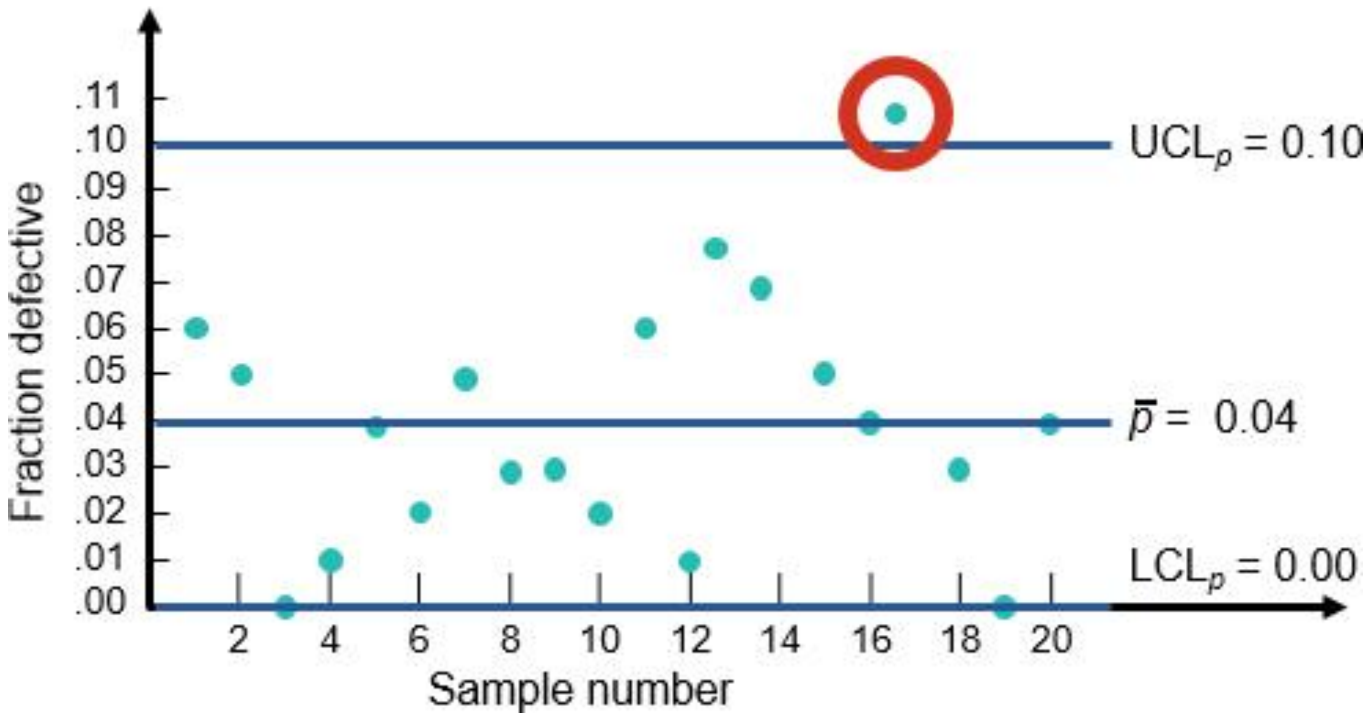
$$\hat{\sigma}_p = \sqrt{\frac{(.04)(1-.04)}{100}} = .02 \text{ (rounded up from .0196)}$$

$$\text{UCL}_p = \bar{p} + z\hat{\sigma}_p = .04 + 3(.02) = .10$$

$$\text{LCL}_p = \bar{p} - z\hat{\sigma}_p = .04 - 3(.02) = 0$$

(because we cannot have a negative percent defective)

p-Chart for Data Entry



Possible assignable causes present

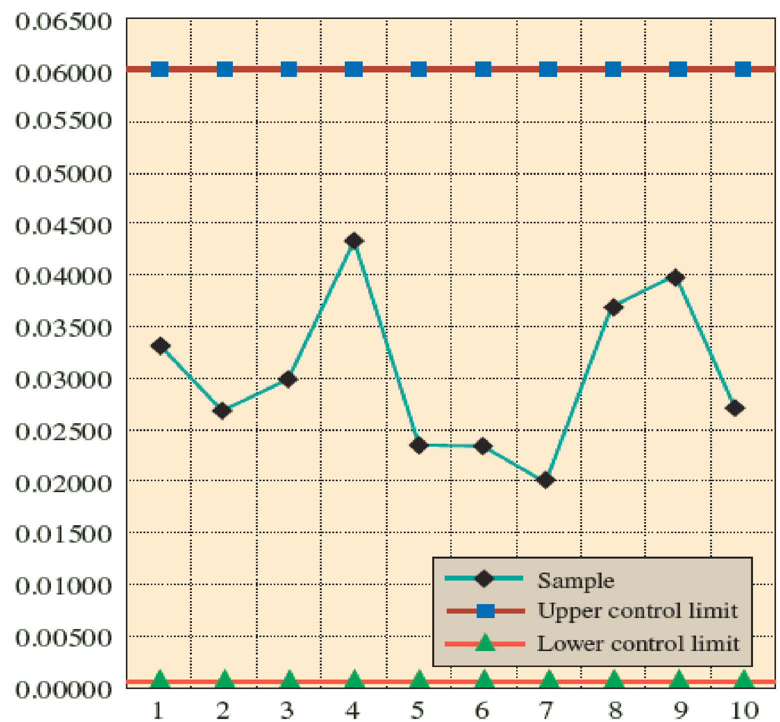
Example: Insurance company

An insurance company wants to design a control chart to monitor whether insurance claim forms are being completed correctly. The company intends to use the chart to see if improvements in the design of the form are effective. To start the process the company collected data on the number of incorrectly completed claim forms over the past 10 days. The insurance company processes thousands of these forms each day, and due to the high cost of inspecting each form, only a small representative sample of 300 was collected each day. The number. of incorrectly filled forms over the 10-day period are 10, 8 , 9, 13, 7, 7, 6, 11, 12, 8. Comment on the effectiveness of the new design.

Example: Insurance company

- $\bar{p} = \frac{91}{10\times300} = 0.03033$
- $s_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = \sqrt{\frac{0.03033(1-0.03033)}{300}} = 0.00990$
- $UCL = \bar{p} + 3s_p = 0.03033 + 3(0.00990) = 0.009900$
- $LCL = \bar{p} - 3s_p = 0.03033 - 3(0.00990) = 0.00063$

Sample	Number Inspected	Number of Forms Completed Incorrectly	Fraction Defective
1	300	10	0.03333
2	300	8	0.02667
3	300	9	0.03000
4	300	13	0.04333
5	300	7	0.02333
6	300	7	0.02333
7	300	6	0.02000
8	300	11	0.03667
9	300	12	0.04000
10	300	8	0.02667
Totals	3,000	91	0.03033
Sample standard deviation			0.00990



Control Limits for *c*-Charts

Population will be a Poisson distribution, but applying the central limit theorem allows us to assume a normal distribution for the sample statistics

$\bar{c}$ = mean number of defects per unit

$\sqrt{\bar{c}}$ = standard deviation of defects per unit

$$\text{Control limits (99.73\%)} = \bar{c} \pm 3\sqrt{\bar{c}}$$

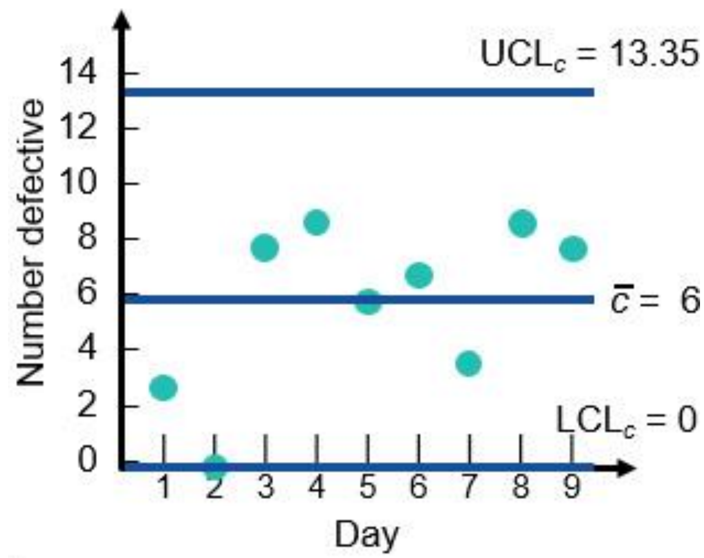
Example: *c*-Chart for Cab Company

Complaints about the behavior of the drivers is recorded. No. of complaints over a 9-day period: 3, 0, 8, 9, 6, 7, 4, 9, 8

$$\bar{c} = 54 \text{ complaints} / 9 \text{ days} = 6 \text{ complaints} / \text{day}$$

$$\begin{aligned} \text{UCL}_c &= \bar{c} + 3\sqrt{\bar{c}} \\ &= 6 + 3\sqrt{6} \\ &= 13.35 \end{aligned}$$

$$\begin{aligned} \text{LCL}_c &= \bar{c} - 3\sqrt{\bar{c}} \\ &= 6 - 3\sqrt{6} \\ &= 0 \end{aligned}$$



Cannot be a negative number

Managerial Issues and Control Charts

Three major management decisions:

- Select points in the processes that need SPC : "which parts of the job are critical to success"
- Determine the appropriate charting technique: "variable or attribute"
- Set clear and specific SPC policies and procedures

Which Control Chart to Use

Variable Data Using an $\bar{x}$- Chart and *R*-chart

1. Observations are **variables**
2. Collect 20 – 25 samples of n = 4, or n = 5, or more, each from a stable process and compute the mean for the $\bar{X}$ -chart and range for the R-chart
3. Track samples of n observations

Which Control Chart to Use

Attribute Data Using a *P*-Chart

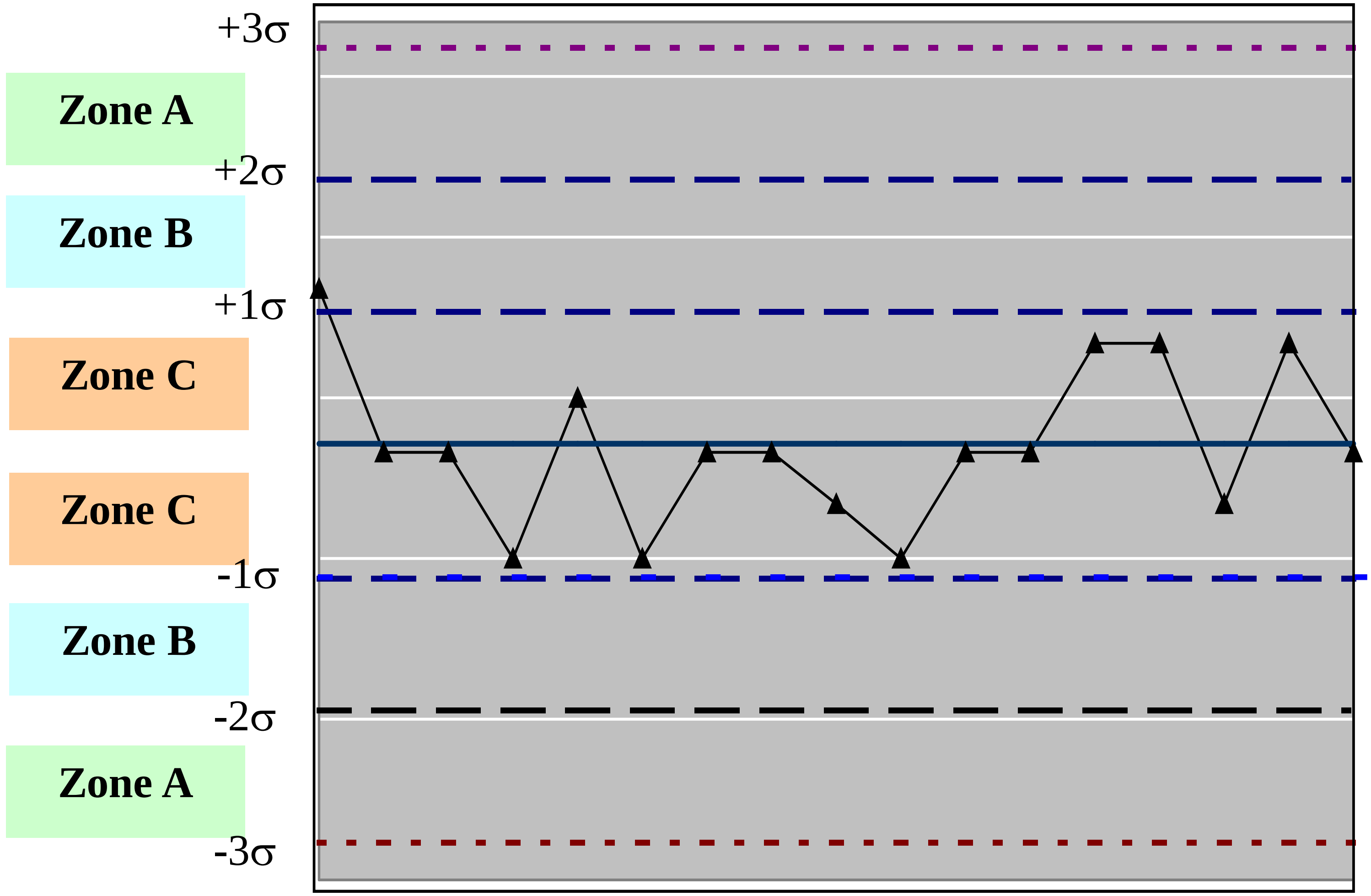
1. Observations are **attributes** that can be categorized as good or bad (or pass–fail, or functional–broken), that is, in two states
2. We deal with fraction, proportion, or percent defectives
3. There are several samples, with many observations in each

Attribute Data Using a *C*-Chart

1. Observations are **attributes** whose defects per unit of output can be counted
2. We deal with the number counted, which is a small part of the possible occurrences
3. Defects may be: number of blemishes on a desk; flaws in a bolt of cloth; crimes in a year; broken seats in a stadium; typos in a chapter of this text; or complaints in a day

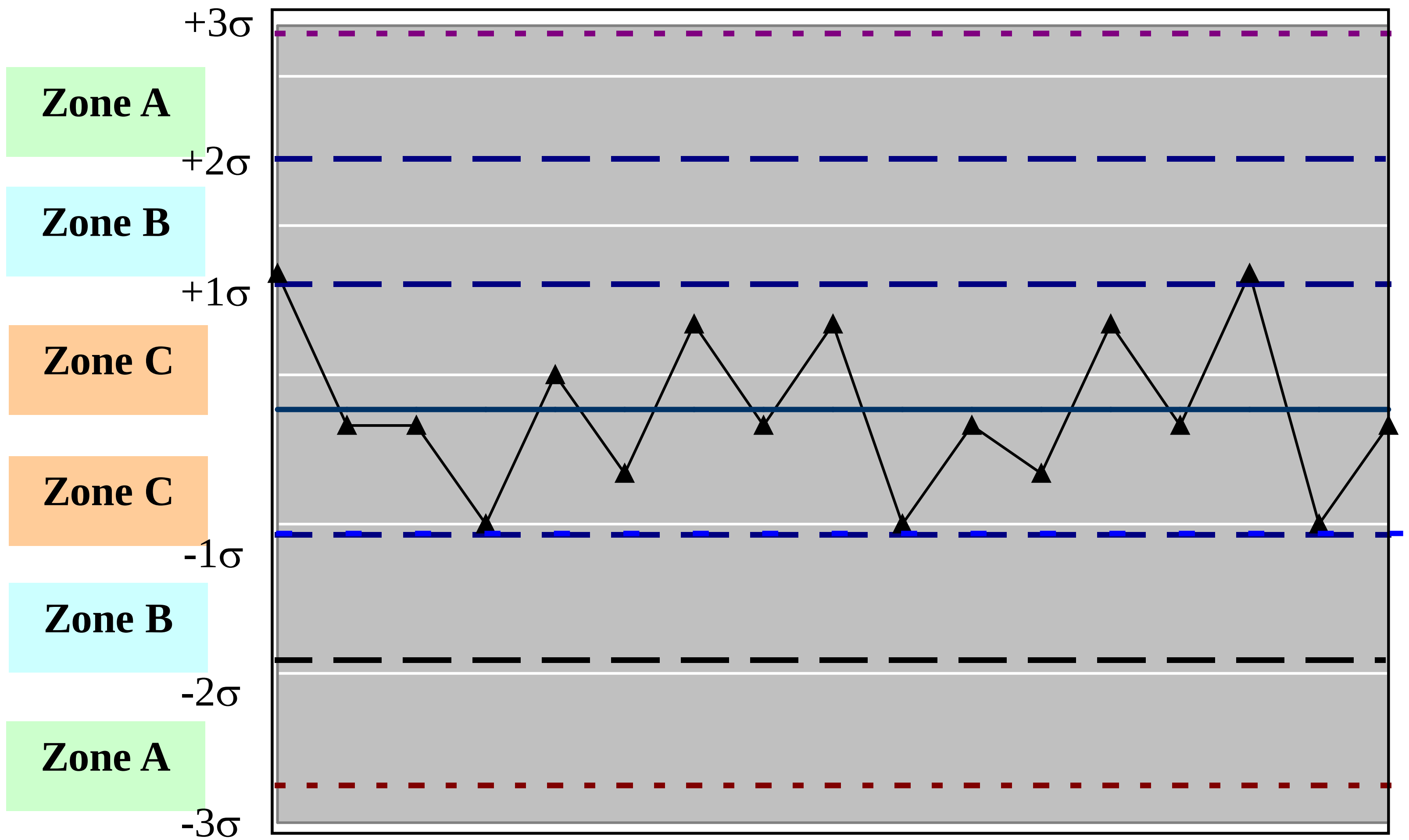
When to stop the process?

15 points in a row in Zone C



When to stop the process?

14 points in a row alternating up and down



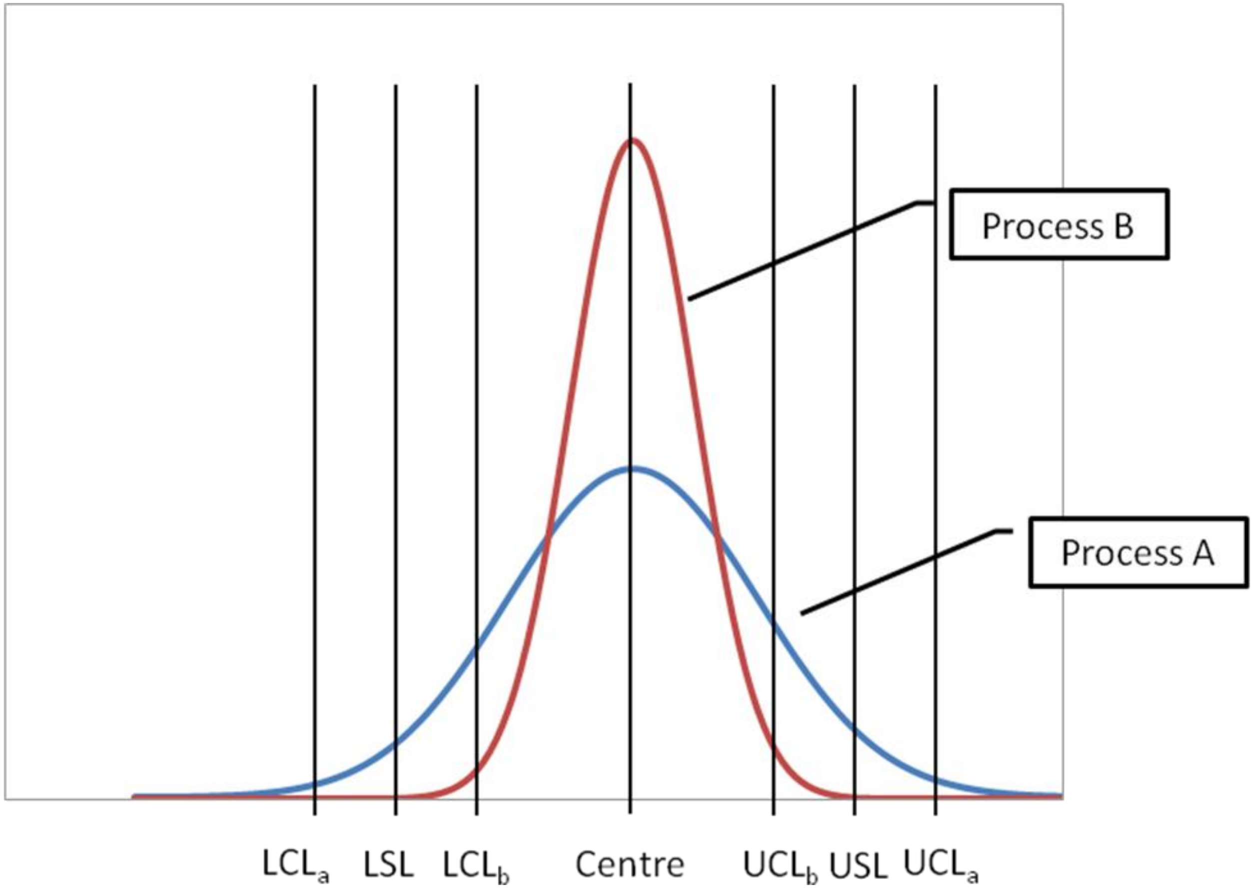
When to stop the process?
Additional rules

- One point beyond Zone A
- Nine points one one side of center line in a row in Zone C or beyond
- Six points in a row, steadily increasing or decreasing
- Fourteen points in a row, alternating up and down
- Two out of three points in a row in Zone A or beyond
- Four out of five points in a row in Zone B and beyond
- Fifteen points in a row in Zone C

Predictive capability of processes

Which process is better?

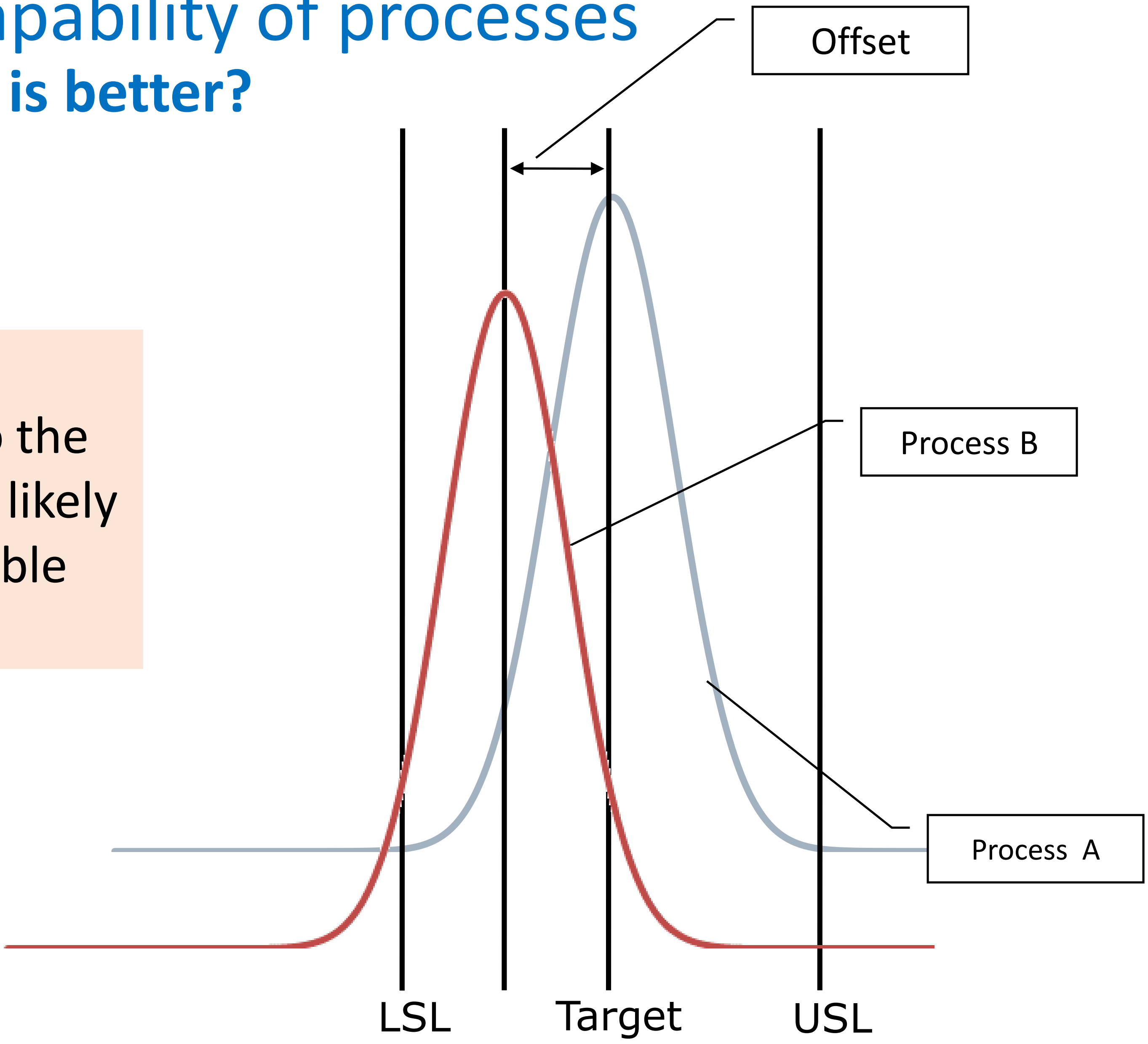
Spread of a process is indicative of its capability



Predictive capability of processes

Which process is better?

A process that is aligned closer to the desired target is likely to be more capable



Process Capability

- The natural variation of a process should be small enough to produce products that meet the standards required
- A process in statistical control does not necessarily meet the design specifications
- **Process capability** is a measure of the relationship between the natural variation of the process and the design specifications

Process Capability

- Process Capability is defined by the spread of the process
- Potential capability (C_p) is defined as the ratio of the difference in specification limits to the process spread

$$C_p = \frac{Specification\ Range}{\mathrm{Pr}ocess\ Capability} = \frac{(USL - LSL)}{6\sigma}$$

- Actual capability (C_{pk}) takes into consideration the extent to which the process has deviated from the desired target

$$C_{pk} = Min\left(\left[\frac{\mathrm{Pr}ocess\ Centre - LSL}{3\sigma}\right], \left[\frac{USL - \mathrm{Pr}ocess\ Centre}{3\sigma}\right]\right)$$

Process Capability Ratio

$$C_P = \frac{\text{Upper Specification} - \text{Lower Specification}}{6\sigma}$$

- A capable process must have a C_p of at least 1.0
- Does not look at how well the process is centered in the specification range
- Often a target value of C_p = 1.33 is used to allow for off-center processes
- Six Sigma quality requires a C_p = 2.0

Process Capability Ratio

Insurance claims process

Process mean $\bar{x} = 210.0 \text{ minutes}$

Process standard deviation $\sigma = .516 \text{ minutes}$

Design specification $= 210 \pm 3 \text{ minutes}$

$$C_p = \frac{\text{Upper Specification} - \text{Lower Specification}}{6\sigma}$$

Process Capability Ratio

Insurance claims process

Process mean $\bar{x} = 210.0$ minutes

Process standard deviation $\sigma = .516$ minutes

Design specification $= 210 \pm 3$ minutes

$$C_p = \frac{\text{Upper Specification} - \text{Lower Specification}}{6\sigma}$$

$$= \frac{213 - 207}{6(.516)} = 1.938$$

Process is capable

Process Capability Index

$$C_{pk} = \text{minimum of} \left(\frac{\begin{matrix}\text{Upper} \\ \text{Specification} - \bar{x} \\ \text{Limit}\end{matrix}}{3\sigma}, \frac{\begin{matrix}\text{Lower} \\ \bar{x} - \text{Specification} \\ \text{Limit}\end{matrix}}{3\sigma} \right)$$

- A capable process must have a C_{pk} of at least 1.0
- A capable process is not necessarily in the center of the specification, but it falls within the specification limit at both extremes

Process Capability Index

New Cutting Machine

New process mean $\bar{x} = .250$ inches

Process standard deviation $\sigma = .0005$ inches

Upper Specification Limit = .251 inches

Lower Specification Limit = .249 inches

$$C_{pk} = \text{minimum of} \left(\frac{(.251) - .250}{(3).0005} \right)$$

Process Capability Index

New Cutting Machine

New process mean $\bar{x} = .250 \text{ inches}$

Process standard deviation $\sigma = .0005 \text{ inches}$

Upper Specification Limit = .251 inches

Lower Specification Limit = .249 inches

$$C_{pk} = \text{minimum of} \left(\frac{(.251) - .250}{(3).0005} \right), \left(\frac{.250 - (.249)}{(3).0005} \right)$$

Process Capability Index

Both calculations result in

$$C_{pk} = \frac{.001}{.0015} = 0.67$$

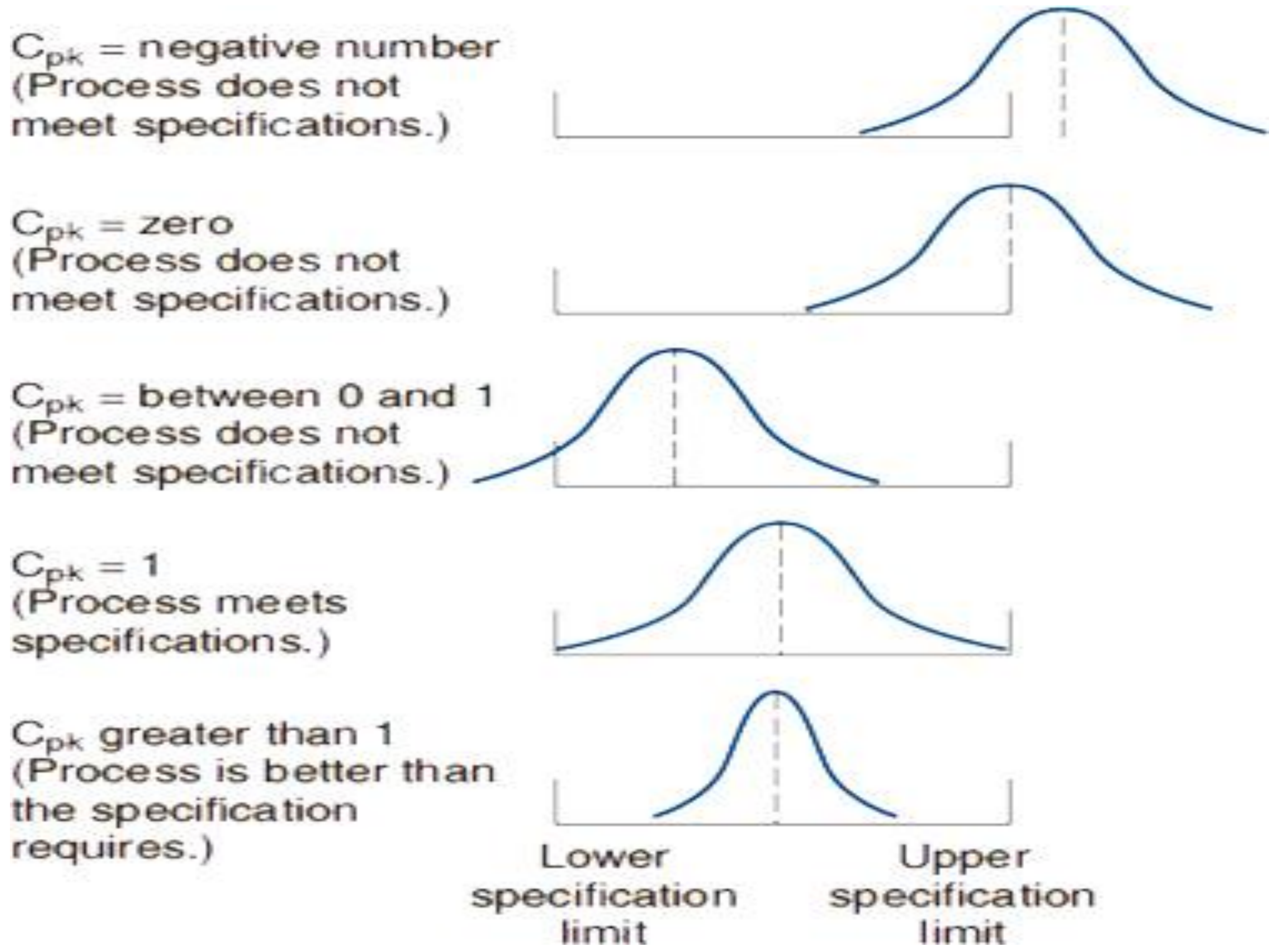
New machine is NOT capable

Process Capability & Defects

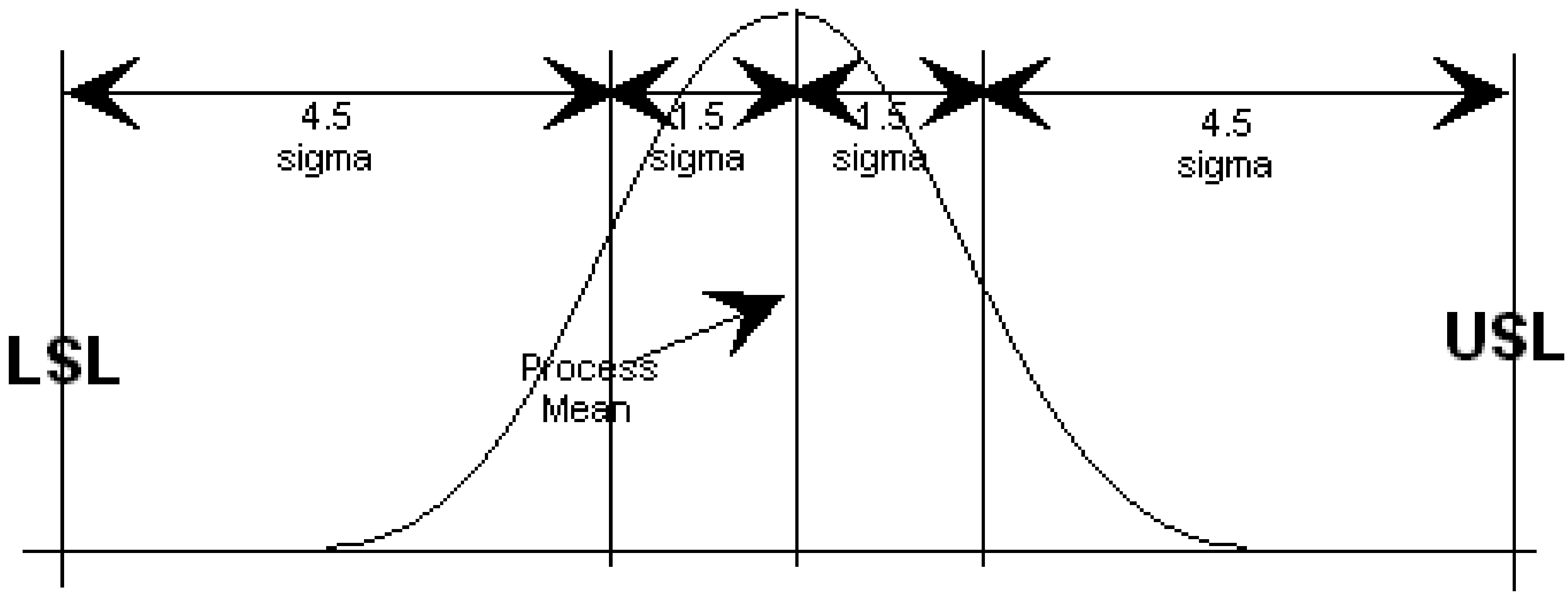
Process Capability Index (C_{pk})	Total Products outside two-sided specification limits
0.25	453,255 ppm
0.50	133,614 ppm
0.60	71,861 ppm
0.80	16,395 ppm
1.00	2,700 ppm
1.20	318 ppm
1.50	7 ppm
1.70	0.34 ppm
2.00	0.0018 ppm

Interpreting C_{pk}

Meanings of C_{pk} Measures



Six sigma quality
A graphical representation



$$C_p = \frac{(USL - LSL)}{6\sigma} = 2 \Rightarrow (USL - LSL) = 12\sigma \Rightarrow A\ spread\ of\ \pm 6\sigma$$